

LAABID Abdessamad

Data Scientist

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SUMMARY

Recent Data Science graduate with a Master's degree and a solid foundation in machine learning and deep learning. Through academic and personal projects, I have developed practical knowledge of Python, scikit-learn, XGBoost, neural networks, CNNs, RNNs, and LSTMs, as well as ML application development and deployment using tools such as FastAPI, Docker, AWS, and CI/CD. I am ready to collaborate with data science and machine learning teams, contribute to real-world projects, learn from experienced professionals, and continue developing my technical skills in a professional environment.

SKILLS

Languages & Libraries: Python, pandas, NumPy, scikit-learn, XGBoost, PyTorch, PyTorch Lightning

Deep Learning: Neural networks, CNNs, RNNs, LSTMs, xLSTM, Transformers (TFT), time series classification

Model Evaluation & Explainability: Cross-validation, Optuna, SHAP, Fairlearn, cost-sensitive evaluation, bootstrap confidence intervals

Data Engineering & MLOps: Pandera, DVC, MLflow, Evidently, pytest, GitHub Actions

Deployment & Cloud: FastAPI, Streamlit, Docker, AWS (Fargate, S3, ECR), AWS CDK, Terraform

EDUCATION

Master's Degree in Data Science and Analytics

July 2026

Cadi Ayyad University – Department of Mathematics and Computer Science

Thesis: "A Multi-Asset Deep Learning Architecture for Continuous Trading Signal Prediction," supervised by Professor Essalih Mohamed. Trained and evaluated 8 deep learning architectures across a 70-stock universe under 5 class-imbalance mitigation strategies (41 trained configurations), evaluated jointly on classification metrics and Sharpe ratio.

PROJECTS

CreditLens – Cost-Sensitive Credit Risk Scoring

Python, scikit-learn, XGBoost, Optuna, SHAP, Fairlearn, Pandera, DVC, MLflow, Evidently, FastAPI, Streamlit, Docker

- Built an end-to-end credit-risk scoring pipeline on the German Credit dataset, optimizing for expected cost (5:1 cost matrix) instead of accuracy
- Calibrated model probabilities and selected a cost-optimal decision threshold using out-of-fold predictions, achieving a test ROC-AUC of 0.76 (95% CI 0.68–0.83)
- Implemented per-applicant SHAP explanations, audited the model with Fairlearn (excluding sex and age at the API boundary), and monitored data drift with Evidently

Car Insurance Claim Risk – Classification & Regression Pipeline

Python, scikit-learn, XGBoost, FastAPI, Docker, AWS (Fargate, ECR, S3), AWS CDK, Terraform, GitHub Actions

- Built an end-to-end project with two scikit-learn Pipelines, predicting claim probability and claim amount
- Reached 0.82 ROC-AUC on claim-probability classification; identified that the regression model did not outperform a naive mean-value baseline and reported this limitation transparently
- Deployed as a containerized FastAPI service on AWS Fargate (provisioned with CDK), with GitHub Actions CI/CD using OIDC-based AWS authentication — no long-lived credentials

Trading Signal Prediction – Multi-Asset Deep Learning (Master's Thesis)

Python, PyTorch, PyTorch Lightning, Darts, TA-Lib, scikit-learn

- Designed and ran a comparative study of 8 deep learning architectures, trained jointly across 70 NASDAQ-100 tickers, under 5 class-imbalance mitigation strategies (41 trained configurations)
- Found the configuration maximizing classification F1 (weighted cross-entropy) also produced the weakest risk-adjusted return, inflating trade count 4–8x
- Identified and corrected a look-ahead leakage issue in a convolutional baseline by implementing and benchmarking a causal variant